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Case report

An open dataset of data lineage graphs for data governance research

Yunpeng Chen [a], Ying Zhao [a], Xuanjing Li [a], Jiang Zhang [b], Jiang Long [b], Fangfang Zhou [a,*]

[a] Central South University, Changsha, Hunan, China
[b] Huawei Cloud Computing Technology Co., Ltd., Hangzhou, Zhejiang, China

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ABSTRACT

Data have become valuable assets for enterprises. Data governance aims to manage and reuse data assets, facilitating enterprise management and enabling product innovations. A data lineage graph (DLG) is an abstracted collection of data assets and their data lineages in data governance. Analyzing DLGs can provide rich data insights for data governance. However, the progress of data governance technologies is hindered by the shortage of available open datasets for DLGs. This paper introduces an open dataset of DLGs, including the DLG model, the dataset construction process, and applied areas. This real-world dataset is sourced from Huawei Cloud Computing Technology Company Limited, which contains 18 DLGs with three types of data assets and two types of relations. To the best of our knowledge, this dataset is the first open dataset of DLGs for data governance. This dataset can also support the development of other application areas, such as graph analytics and visualization.

1. Introduction

Data are newly emerged assets for enterprises (Telenti and Jiang, 2010; Birch et al., 2021). Data governance refers to the use of various policies, standards, and techniques to promote the availability, quality, and security of data assets for enterprise management and product innovations (Abraham et al., 2019; Janssen et al., 2020). Currently, most data assets exist in the form of data tables. These tables commonly present complex relationships established by application-driven data transformations between tables. These relationships are known as data lineages (Patel et al., 2020; Tang et al., 2019) and are represented by data asset graphs. Fig. 1 demonstrates a data lineage graph (DLG) in which a node represents a data asset and an edge represents a relation between two data assets. The table node T1 records the sale data of a certain enterprise's products. To enhance product sales, the data job node J1 uses a set of Structured Query Language (SQL) codes to extract consumer information (e.g., age, gender, and occupation) from T1 and then save it into the table node T2. The data job nodes J2, J3, and J4 use SQL codes to extract the statistical information, including purchase, retention, and churn rates, from T1 and then save it into the table node T3.

DLGs provide enterprises with a bird's-eye view of entire data assets and their processing life cycle, thereby supporting a wide range of crucial requirements of data governance. For example, DLGs can depict complex relationships among data assets, allowing data users to trace the path of data transformation or data flow. The data in T2 and T3 in Fig. 1 originate from T1, as evidenced by the directions of data transformations in the DLG. DLGs can help data users find core data assets. In Fig. 1, T1 is a core data asset because it serves as a base table for generating new tables. T2 and T3 are potential core data assets because they would be applied in product sale optimizations. Therefore, data governance techniques based on data lineage analysis have recently received considerable attention from industry and academia (Tang et al., 2019; Freche et al., 2021).

The progress and application of methods and techniques in a specific field rely heavily on the availability of high-quality open datasets, such as the ITD-2018 dataset (Zhao et al., 2022b) in insider threat detection, the malicious webshell family (Zhao et al., 2023b) dataset in deep learning, and the ICMTD-2019 dataset (Zhao et al., 2021b) in the indoor navigation and crowd behavior monitoring. However, few DLG datasets currently are open to facilitate the research and development of relevant methods and techniques for data governance. The main reason possibly is that DLGs contain rich sensitive business information. Enterprises are commonly reluctant to release such datasets publicly because of privacy and security concerns.

This paper introduces an open dataset of Data Lineage Graphs for Data Governance Research (DLG-DG-23). This dataset is a real-world dataset sourced from Huawei Cloud Computing Technology Company Limited (referred to as "Huawei Cloud"), which is the fifth-largest global cloud service provider (Gartner Research, 2020) . The raw data in the dataset have been subjected

* Corresponding author.
E-mail address: zff@csu.edu.cn (F. Zhou).

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Table 1
Descriptions of data asset types and relation types.

	Type	Description
Data Asset	Data Table Data Job Data Field	A relational data table that stores information A segment of SQL codes used to process data in data table assets A column in a data table asset
Relation	DATA_FLOW PARENT_CHILD	Data transformation between data table assets and data job assets Affiliation between data table assets and data field assets

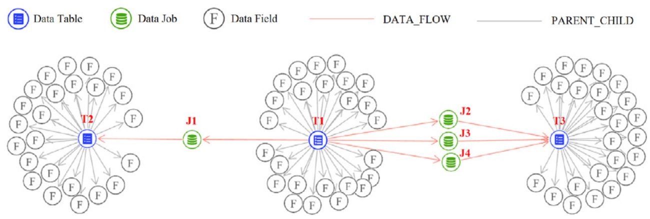

Fig. 1. Example of a DLG that contains 81 datasets and 82 relations.

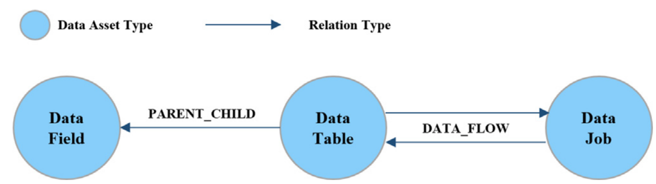

Fig. 2. Illustration of a heterogeneous graph model of the DLG-DG-23 dataset.

to a set of data cleansing and rigorous anonymization processes to ensure high quality and eliminate privacy and security risks. The dataset contains 18 DLGs with diverse sizes from 278 nodes to 17,085 nodes. These DLGs originate from three application scenarios in data governance. Each DLG contains the three most common types of data assets (i.e., data table, data job, and data field) and two types of relations (i.e., DATA_FLOW and PARENT_CHILD), which can be used to describe extract, transform, and load (ETL) processes (Vassiliadis et al., 2002) in application scenarios. To the best of our knowledge, DLG-DG-23 is the first open DLG dataset in the field of data governance. The dataset can promote the development of methods and techniques in the field of data governance. Moreover, DLGs are directed, sparse, heterogeneous, and scale-free graphs, making them suitable for a wide range of research in the field of graph analytics and visualization.

2. Dataset description

2.1. Abstracted DLG model

The DLG-DG-23 dataset can be abstracted through a heterogeneous graph model, as illustrated in Fig. 2. This model consists of three types of data assets and two relation types. As described in Table 1, data table assets are the main form of data assets. Data job assets typically consist of SQL codes that extract and manipulate data in data table assets, constructing data transformations among data table assets. Data field assets are the fundamental units for storing and organizing data within data table assets.

DATA_FLOW relations represent the directions of data transformation between data table assets and data job assets. In Fig. 1, a DATA_FLOW relation from the data table asset T1 to the data job asset J1 signifies that the information in T1 is extracted and manipulated by J1. Likewise, a DATA_FLOW relation from the data job asset J1 to the data table asset T2 implies that the processed information is loaded into T2 by J1. PARENT_CHILD relations represent the affiliations between data table assets and data field assets. In Fig. 1, the data table asset T1 has many PARENT_CHILD relations, indicating that these data field assets are constituent parts of T1.

2.2. Dataset construction pipeline

This section briefly introduces the construction pipeline of the DLG-DG-23 dataset. We invited data experts from Huawei Cloud to participate in the dataset construction. Three criteria were considered in the dataset construction. (1) The dataset should consist of multiple DLGs. (2) The DLGs should cover diverse application scenarios. (3) The DLGs should present diverse complexity and sizes.

Eighteen DLGs were selected based on the above three criteria and were sourced from three application scenarios that require data governance, namely, cloud infrastructure, customer service, and operation analysis. The cloud infrastructure aims to provide reliable and efficient foundations, enabling enterprises to deploy, manage, and deliver cloud services. The customer service aims to increase customer satisfaction and foster retention. The operation analysis conducts in-depth research and evaluation of enterprises' operational processes, thereby obtaining crucial business insights for decision-making. These application scenarios frequently appear in enterprise management. Then, the three most common data asset types and two relation types were preserved in the dataset, which can fully describe ETL processes that are commonly used in data integration and warehousing. Moreover, anonymization operations were performed on the names of three data assets and two relations to prevent the disclosure of private information. The SQL codes within data job assets were discarded to avoid copyright issues. Data cleansing and inspection steps were conducted to ensure the information integrity for each data asset and relation.

2.3. Dataset information

The DLG-DG-23 dataset is stored in JSON format with a total uncompressed volume of 24 MB. It consists of 18 connected DLGs, including 10 small-scale DLGs with approximately hundreds of data assets, 6 medium-scale DLGs with around thousands of data assets, and 2 large-scale DLGs with more than 10,000 data assets, as shown in Table 2. Each DLG is a strongly connected graph (Xin et al., 2019). Three DLGs with different sizes are visualized using node-link diagrams in Fig. 3.

A DLG is stored in a Node.json file and an Edge.json file. The Node.json file has two fields to provide the information of a data asset, as shown in Table 3. The Edge.json file has four fields to provide the information of a relation, as shown in Table 4.

3. Applied areas of dataset

This section introduces the potential application areas of the DLG-DG-23 dataset.

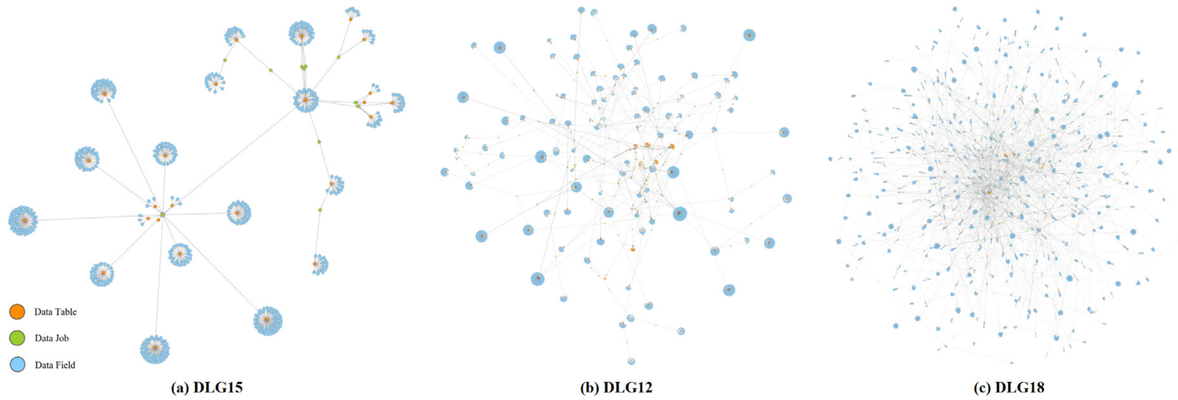

Fig. 3. DLG visualizations including (a) a small-scale DLG with 558 data assets and 564 relations, (b) a medium-scale DLG with 5,645 data assets and 5,802 relations, and (c) a large-scale DLG with 17,085 data assets and 17,720 relations.

Table 2
Basic information of DLGs in the DLG-DG-23 dataset.

Application scenarios	ID	Nodes	Edges	Scale
Cloud Infrastructure	DLG1	298	298	small
	DLG2	464	467	small
	DLG3	603	610	small
	DLG4	415	414	small
	DLG5	1,299	1,351	Medium
	DLG6	13,840	14,325	Large
Customer Service	DLG7	574	574	small
	DLG8	546	548	small
	DLG9	756	771	small
	DLG10	5,378	5,555	Medium
	DLG11	2,024	2,032	Medium
	DLG12	5,645	5,802	Medium
Operation Analysis	DLG13	2,157	2,211	Medium
	DLG14	453	452	small
	DLG15	558	564	small
	DLG16	5,574	5,876	Medium
	DLG17	651	652	small
	DLG18	17,085	17,720	Large

Table 3
Field descriptions of Node.json file.

Field	Description	Format	Example
asset_id	The unique identification of a data asset	String	"7c036def-1482-4ed2 -85b6-a761da4268cc"
asset_type	The type of data asset	String	"Data Table", "Data Job", or "Data Field"

Table 4
Field descriptions of Edge.json file.

Field	Description	Format	Example
relation_id	The unique identification of a relation	String	"7fdeb419-69e1-4743 -9522-63a568e530fb"
relation_type	The type of a relation	String	"DATA_FLOW" or "PARENT_CHILD"

3.1. Data governance

Data governance is the primary application area of the DLG-DG-23 dataset, and its data users include data stewards, data experts, data engineers, and data analysts. The dataset can provide multi-perspective insights for various data users in data governance. Errors may occur in data transformations because of data quality issues, incompatible data integration, and flawed transformation logic. These errors are known as data anomalies (Homayouni, 2018). Quickly tracing back and identifying data assets responsible for anomalies is crucial to prevent the propagation of erroneous data. DLGs provide an overview of data assets and relevant data lineage, thus helping data stewards gain deeper insights into data accuracy and consistency by examining data transformations. This approach enables them to efficiently locate anomalous data assets. Anomaly detection techniques such as outlier node detection (Boukerche et al., 2020) and anomaly pattern detection (Zhao et al., 2023a) can be applied to trace anomalous data sources in DLGs.

A set of core data assets may exist in a DLG. Generally, core data assets are high-value data assets that may play critical roles in enterprise management or product innovation. For example, the data table asset T1 in Fig. 1 is a core data asset because it provides important data support in data transformations. Core data asset identification reminds data experts which data assets have the potential to empower management and product innovation. Diverse graph analysis techniques, such as node centrality measurement (Rodrigues, 2019) and node role identification (Cheng et al., 2022), can be applied for core data asset identification. Moreover, business knowledge is important in core data asset identification (Chen et al., 2023). We invited data experts from Huawei Cloud to manually label the core data assets of six DLGs based on their personal experience and professional knowledge. The six DLGs and their core data assets IDs are listed in Table 5. The labels provide a reference significance to promote the development of core data asset identification research.

In data governance, alterations in data assets may lead to unknown potential influences for subsequent data assets during data transformations. Data engineers should concentrate on monitoring alterations in particular data assets to comprehend their impacts on both global and local levels. This allows them to adeptly control and adjust the extent and intensity of these impacts. DLGs represent data assets as nodes and connect them through one-way edges, enabling data engineers to monitor the data flow and estimate the potential influence of data changes for the entire graph. Dynamic graph analysis (Burch et al., 2021) and sensitivity analysis (Murai and Yoshida, 2019) can help data engineers analyze the influence of data changes on data transformations in DLGs.

Data transformations between data assets in data governance reflect specific business requirements. Data analysts can obtain valuable insights from data transformations to support future business decisions. By observing and analyzing the attributes of nodes and edges in DLGs, data users can discover patterns, trends, and regularities among data assets, thus assisting decision-makers in understanding the development status and

Table 5
Core data assets in six DLGs.

DLG ID	Core data asset IDs
DLG1	35152af4-9211-4a5e-abca-feb49c70743d 4d7895e6-6376-401d-9164-792eb4327195 c55e1828-6b17-4bbb-b7b5-03d3c794eddd
DLG2	d7a8326b-460e-4238-b4ce-c95a964adf9a 783ba88f-86a9-463d-a329-7be8a122a452 c3e4a4ba-9ad4-4364-8e65-3baaa76a0481 65fd86fb-9321-4b16-9888-d7afd62202b0 4b0b3d72-4602-440a-b190-75859501c1d5
DLG9	2d975164-2e40-4331-8e79-89562690dad0 cc80bf0f-7e4f-44a1-9b6e-976755b48929 0c3aff84-25dd-4a10-bc0b-86e59ef21770 3a71edf0-e416-4064-b2c5-fa04c90acea1 26ef0607-467f-481e-b284-868f4bee1c43 02af7d6f-f5ab-45fa-8e40-f91a3dcd3c78 7d171f49-c15c-44df-8a9e-88fd45b322ac 9d7beaa4-fe0b-417b-ba0c-8040107e7873
DLG11	70d8371f-9262-4139-aae4-55f6321acbbd 29133f78-aa7e-4d97-a578-6efef757f985 67d79363-c4d5-4776-9e86-a4bc246dda9b 432f5499-a617-40d5-b647-777ff04e09cb fb73ddcc-ae25-492c-9fca-1a3e9534459f 6fb8299c-ffe8-47c2-87e2-e01d5ea06029
DLG13	fb73ddcc-ae25-492c-9fca-1a3e9534459f 29133f78-aa7e-4d97-a578-6efef757f985 644390eb-5d14-46f8-8c85-0a0b74bdfc52 9b408b3d-c8b2-4c99-a224-34463610d30d 62633e2c-2819-467d-9bf2-8e85bde00baf ffd28ea3-3784-45b2-87fa-99214ba15137 055d4918-ce84-4fe4-957b-b044dbff9785 0382984d-09f9-47f2-bbac-0eb252d6723d 86245453-d208-4260-8c8a-8ed299d1bb51 a08abfe2-9210-4936-893e-29852aedcaf4 cab58466-d666-4e4c-be99-209e552fc11d
DLG15	0c3aff84-25dd-4a10-bc0b-86e59ef21770 a24915a6-f08e-4990-9c01-91a83b5d2cb6 7e3ec6e6-7735-4276-af43-96f75586f247

future trends of a certain business event related to these data assets. Data exploration (Leite et al., 2020; Wang et al., 2023) and knowledge discovery (Zhou et al., 2023) can be used for decision support in DLGs, providing deeper insights and recommendations for future business decisions.

3.2. Graph analytics and visualization

DLG-DG-23 dataset is suitable for many research branches of graph theory (Riaz and Ali, 2011; Zhao et al., 2022a; Fischer et al., 2020). DLGs in the dataset are sparsely directed graphs because the edges are directed and the ratio of nodes to edges in DLGs is nearly 1:1. Sparse graphs are commonly used in node centrality analysis and compressive sensing. DLGs are heterogeneous graphs that have many node types and edge types. Heterogeneous graphs are widely used in knowledge graph construction and application. Rich cluster structures and diverse bridges are present between clusters in a DLG, thereby aiding community detection, node importance identification, and critical path mining. DLGs are scale-free graphs because the degrees of all nodes present a power-law distribution. Scale-free graphs are widely used in hotspot analysis, propagation analysis, and infectious disease prediction.

DLGs bring opportunities and challenges for various graph visualization techniques (Zhao et al., 2021a; Zhou et al., 2021; Peng et al., 2023). First, the DLG-DG-23 dataset examines the performance of graph layout methods. A large DLG generally has more than 10,000 nodes and edges, thereby posing a challenge for existing graph layout methods in obtaining a clear layout. Second, a DLG contains numerous notable structures, thereby facilitating the exploration of advanced graph simplification and abstraction techniques such as edge binding and graph sampling to comprehensively illustrate these structures through graph visualization.

4. Conclusion

This paper presents an open dataset of data asset graphs for data governance research. First, this paper describes the heterogeneous data lineage graph model, which includes three types of nodes and two types of edges. Then, this paper introduces the dataset construction pipeline and the basic information of the DLGs in the dataset. Finally, this paper summarizes the potential application areas of the dataset. We expect that the dataset can facilitate the development of application areas in data governance. Moreover, we expect that the dataset can contribute to other application areas such as graph analytics and visualization.

The future work includes two main aspects. First, we aim to improve the universality of the DLG-DG-23 dataset. This dataset is constructed by abstracting data assets and their lineage relationships within three typical business scenarios from Huawei Cloud. More business scenarios from Huawei Cloud and other enterprises should be introduced to enhance its universality. Then, we intend to enrich the information within the DLG-DG-23 dataset. This dataset provides only a few information fields for nodes and links and only one label type (i.e., core data asset). We will collaborate with data asset experts to augment the dataset with rich node and edge information and various labels.

Ethical Approval

This study does not contain any studies with human or animal subjects performed by any of the authors.

CRediT authorship contribution statement

Yunpeng Chen: Investigation, Writing – original draft preparation. **Ying Zhao:** Writing – reviewing and editing. **Xuanjing Li:** Data curation. **Jiang Zhang:** Resources. **Jiang Long:** Validation. **Fangfang Zhou:** Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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